MICHAEL ADELEKE

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RESEARCH INTERESTS

Radar-based sensing and micro-Doppler signal processing; machine learning for interactive systems; human-computer interaction; equitable and accessible AI/ML education; virtual reality for learning and empathy; broadening participation in computing.

Teaching Interests: introductory programming, machine learning, human-computer interaction, and computing education.

EDUCATION

The University of Alabama *Tuscaloosa, AL*
Ph.D. in Computer Science, expected May 2030
Advisor: Dr. Chris Crawford, Human-Technology Interaction Lab. GPA: 4.0

Morgan State University *Baltimore, MD*
B.S. in Computer Science, December 2024
Advisor: Dr. Naja Mack, Human-AI eXperience Lab. Summa Cum Laude. GPA: 3.87

TEACHING AND MENTORING EXPERIENCE

Adjunct Faculty *Jan 2025 – May 2025*
Department of Computer Science, Morgan State University, Baltimore, MD
- Taught Introduction to Computer Science I to 15 students as instructor of record.
- Contributed to curriculum development in computing and emerging technologies.

Program Coordinator, Summer AI Research Institute (SAIRI) *Summers 2025 – 2026*
Center for Equitable AI & Machine Learning Systems (CEAMLS), Morgan State University, Baltimore, MD
- Coordinated and supported 87 undergraduate researchers across two summer cohorts of an AI research program.
- Managed logistics, communication, mentor coordination, and research workflow through final presentations.

Camp Coordinator, CodeBears *Summers 2023 – 2025*
Human-AI eXperience Lab, Morgan State University, Baltimore, MD
- Coordinated a K-12 computing camp for 160+ summer campers across three consecutive summers, introducing programming fundamentals.
- Extended the program to 100+ students through Spring Break and Saturday Academy sessions.
- Reached 2,100+ K-12 students through STEM day outreach events.
- Program outcomes published at SIGCSE 2024 and BICE 2024.

RESEARCH EXPERIENCE

Graduate Research Assistant *Aug 2025 – Present*
Human-Technology Interaction Lab (HTIL), The University of Alabama, Tuscaloosa, AL
- Investigate radar-based sensing systems, analyzing signal behavior and data interpretation to build interactive human-technology experiences.
- Lead development of SensDSv2, a platform that makes micro-Doppler radar sensing usable for hands-on machine learning education (see Research Software).
- Positioned radar sensing as a novel modality for ML and robotics education through a literature review spanning HCI, human-robot interaction, and computing education.

Lead Undergraduate Researcher *Dec 2022 – Aug 2025*
Human-AI eXperience (HAX) Lab, Morgan State University, Baltimore, MD
- Designed and evaluated applied research systems across AI, VR, and STEM education, including VR concussion education tools, conversational AI bias detection, and K-12 STEM outreach programs.
- Contributed to culturally responsive youth computing programs and publications at SIGCSE, CHI, and BICE.

Summer Research Assistant *May 2023 – Aug 2023*
Human-Technology Interaction Lab, The University of Alabama, Tuscaloosa, AL
- Built interactive learning systems integrating robotics, physiological sensors, and machine learning.

PUBLICATIONS

Peer-Reviewed Conference Papers

1. Naja A. Mack, Clyde W. Tandjong, **Michael B. Adeleke**, Elijah Ballou, Amyra Harry, and Jaunel Panton. 2025. Revolutionizing Culturally Relevant Math Education Through AI and Co-Designing Strategies. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26 – May 1, 2025, Yokohama, Japan. ACM, 6 pages. https://doi.org/10.1145/3706599.3719835
2. Naja A. Mack, **Michael B. Adeleke**, Elijah Ballou, Destiny Davis, Vincent Ingram, and Katlyn Cox. 2024. Breaking Stereotypes and Feeding the STEM Pipeline. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2024)*, March 20–23, 2024, Portland, OR, USA. ACM, 7 pages. https://doi.org/10.1145/3626252.3630793
3. Naja A. Mack, **Michael B. Adeleke**, Vincent Ingram, Elijah Ballou, Jayden K. Briggs-Belt, and Antwain Jordan. 2024. CodeBears: Key Insights Gained from a Summer Coding Camp Empowering Underrepresented Youth. In *2024 Black Issues in Computing Education (BICE)*, February 2024. IEEE, 80–86. https://doi.org/10.1109/BICE60192.2024.00021

PRESENTATIONS AND RESEARCH COMPETITIONS

1. *Collision Course: Elevating Concussion Awareness Through Immersive Virtual Reality.* International Conference on Applied Human Factors and Ergonomics (AHFE), Summer 2025. Presenter.
2. *CodeBears: Breaking Stereotypes and Feeding the STEM Pipeline.* ACM Undergraduate Research Competition, ACM Richard Tapia Celebration of Diversity in Computing, 2024. Poster and presentation, 3rd Place.
3. *Collision Course.* Morgan TechFest Poster Competition, 2023. Co-author, 2nd Place.
4. *VR Stress Assessment Using EmotiBit.* STARS Celebration, 2023. Presenter.

RESEARCH SOFTWARE AND SYSTEMS

SensDSv2 *Radar-Based ML Education Platform*

- PyQt6 desktop application supporting live micro-Doppler spectrogram visualization, gesture data collection, Vision Transformer model training, prediction testing, and optional VEX AIM robot control.
- Implemented the full signal processing pipeline (Range FFT, MTI, STFT) for the Infineon BGT60TR13C 60 GHz radar sensor, including chirp parameter configuration through the Infineon SDK.
- Built a PCA-based novelty scoring module that flags classroom-collected captures falling outside the training feature space, motivating on-device training as a design target.
- Optimized model selection and inference scheduling for classroom deployment on Surface Pro hardware.
- Distributed as signed macOS and Windows executables with a user guide; a browser-based Web Serial version is now functional.

radar-js *JavaScript Library, In Development*

- Browser-based JavaScript library, modeled on bci.js, that connects to the Infineon BGT60TR13C radar over Web Serial and streams data in real time with no drivers or local installs.
- Enables cross-platform radar sensing and visualization on macOS, Windows, and other devices directly in the browser.

AWARDS, HONORS, AND FELLOWSHIPS

- SREB Doctoral Fellow, The University of Alabama, 2025 – Present
- Graduate Council Fellow, The University of Alabama, 2025 – Present
- Summa Cum Laude, Morgan State University, 2024
- 3rd Place, ACM Undergraduate Research Competition, Tapia Celebration of Diversity in Computing, 2024
- 2nd Place, Poster Competition, Morgan TechFest, 2023
- Best Technical Solution, Be Smart Hackathon, 2023 and 2024

PROFESSIONAL MEMBERSHIPS

Association for Computing Machinery (ACM), Student Member; National Society of Black Engineers (NSBE), Member

TECHNICAL SKILLS

Programming Languages: Python, Java, C#, JavaScript, Ruby
Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, Vision Transformers, CNNs, RNN/LSTM, PCA
Signal Processing and Sensing: micro-Doppler spectrogram analysis, Range FFT, MTI filtering, STFT, MNE (EEG), EmotiBit
Application Development: Node.js, Web Serial API, JavaScript library development, real-time browser visualization, PyQt6, pyqtgraph, Unity, VR development, PyInstaller, Git and GitHub Actions
Generative AI and LLMs: Claude, Claude Code, ChatGPT, prompt engineering